

Prathamesh Nikam

Machine Learning & Deep Learning Engineer

Hyderabad, India | +91 7249725186 | prathmeshnikam2208@gmail.com



github.com/Prathmesh333



linkedin.com/in/prathamesh-nikam-89b614210



prathmesh333.github.io

SUMMARY

M.Tech IT graduate focused on distributed deep learning, predictive modeling, and NLP. Built a published PyTorch/Ray framework, multi-task dialogue models, and probabilistic forecasting pipelines with benchmark-driven evaluation.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, Ensemble Learning, Feature Engineering, Model Evaluation, Probabilistic Forecasting

Deep Learning: PyTorch, TensorFlow, Transformers, DeBERTa, CNNs, ResNet-18, BiGRU, Multi-Task Learning

Data & Systems: Python, SQL, NumPy, Pandas, Ray, FedAvg, Quantization, FastAPI, Docker, Git

SELECTED PROJECTS

HQDE | Hierarchical Quantized Distributed Ensemble [Link]

Python, PyTorch, Ray, ResNet-18, FedAvg

- Developed and published a distributed deep-learning framework with stateful Ray workers, independent ensemble training, and synchronized FedAvg-style updates.
- Benchmarked five image datasets: independent HQDE exceeded centralized accuracy on four, with a largest observed gain of 2.62 percentage points.
- Implemented blockwise delta quantization with error feedback, reducing model-update communication by approximately 3.05x versus matched FP32 FedAvg.

EvidenceMem | Visual Memory and Continual Classification [Link]

Python, PyTorch, OpenCLIP, FAISS, NumPy, MiniBatchKMeans

- Built a compact medoid memory over frozen CLIP embeddings with visual-text fusion, retrieved image evidence, and new-class insertion without encoder updates.
- Reported 89.58% mean CIFAR-10 accuracy across three seeds with 200 stored prototypes, 4.33 percentage points above zero-shot CLIP.
- Evaluated CIFAR-100 classification, OOD detection, memory budgets, and component ablations; documented the accuracy gap to a trained linear probe.

ForecastForge | Probabilistic Revenue Forecasting [Link]

Python, XGBoost, Empirical Bayes, Monte Carlo Simulation, Streamlit

- Built 30/60/90-day revenue and ROAS forecasts using empirical Bayes priors, cold-start peer transfer, and reconciled P10/P50/P90 intervals.
- Implemented leakage-safe rolling backtests and deterministic scoring; selected XGBoost after it trained 4.5x faster than CatBoost in project benchmarks.

EXPERIENCE

University of Hyderabad | Teaching Assistant - Web Development (IT) Lab | Jan - Jun 2026

- Guided students on project design and full-stack application development; recommended GenAI tools for specific project needs, suggested improvements, and supported project evaluation.

University of Hyderabad | Teaching Assistant - Java OOP Lab | Jul - Dec 2025

- Designed lab questions, evaluated student work, and guided students through Java object-oriented programming concepts and debugging.

EDUCATION

University of Hyderabad | M.Tech, Information Technology | 2024 - 2026 | CGPA: 8.68

Modern Education Society College of Engineering | B.E., Computer Engineering | 2019 - 2023

Honors in AI/ML | B.E. CGPA: 8.85

ACHIEVEMENTS

Data Science Conference Hackathon: Winner (2026) | AI4Bharat AWS Hackathon: Finalist | SuperRR Selection Hackathon: Top 36 of 500+ participants (2025) | Finalist in 10+ hackathons held across India.